ASSIGNMENT

1 The total number of arrangements which can be made out of the letters of the word 'Algebra', without altering the relative position of vowels and consonants is

Solution:

Here, without altering the relative position means, vowels will arrange where vowels are there and consonants will arrange where consonants are there

$\therefore$ Total number of ways = $\dfrac{4!3!}{2!}$

2 (a) If the letters of the word "VARUN" are written in all possible ways and then are arranged as in a dictionary, then the rank of the word VARUN is

(b) The letters of the word RANDOM are written in all possible orders and these words are written out as in a dictionary then the rank of the word RANDOM is

Solution:

(a) Arranging letters in alphabetical order

A, N, R, U, V

A					$\longrightarrow$ 4!
N					$\longrightarrow$ 4!
R					$\longrightarrow$ 4!
U					$\longrightarrow$ 4!
V	A	N			$\longrightarrow$ 2!
V	A	R	N		$\longrightarrow$ 1!
V	A	R	U	N	$\longrightarrow$

$\therefore$ Rank of Varun $= 4(4!) + 2 + 1 + 1$

$= 96 + 4 = 100$

(b) Arranging letters of RANDOM in alphabetical order, we get

A, D, M, N, O, R

Thus before words starting from 'R'

We will have 5(5!) words

Now,

R	A	D				$\longrightarrow 3!$
R	A	M				$\longrightarrow 3!$
R	A	N	D	M		$\longrightarrow 1!$
R	A	N	D	O	M	

$\therefore$ Total words or rank of Random

$=5(5!)+2(3!)+2$

$=600+12+2$

$=614$

3 The total number of ways of arranging the letters AAAA BBB CC D E F in a row such that letters C are separated from one another is

Solution:

Here, $\times A\times A\times A\times A\times B\times B\times B\times D\times E\times F\times$

If C is arranged in these gaps then they are not together

$\therefore$ Number of ways

$={}^{11}C_2\times\frac{10!}{4!3!}$

$=1386000$

4 The total number of permutations of 4 letters that can be made out of the letters of the word EXAMINATION is

Solution:

Given,

2A, 2I, 2N, 1E, 1X, 1M, 1T, 1O

Different permutation are

(i) 2 alike, 2 alike $={}^{3}C_2\times\frac{4!}{2!}=18$

(ii) 2 alike, 2 different $={}^{3}C_1\times{}^{7}C_2\times\frac{4!}{2!}=756$

(iii) all different $={}^{8}C_4\times 4!=1680$

$\therefore$ Total words = 18 + 756 + 1680 = 2454

5 For the number $3^4\times5^2\times7^3$, find

(a) Number of divisors

(b) Number of even divisors

(c) Number of divisors which are multiple of 21

(d) Sum of the divisors

Solution:

(a) Given, $3^4\times5^2\times7^3$

Number of divisors = (4 +1) (2 + 1) (3 + 1)

= 5 × 3 × 4 = 60

(b) Number of even divisors = 0 (as no even digit is there)

(c) Number of divisors which are multiple of 21

$=(4)(2+1)(3)$
$=4\times 3\times 3$
$=36$

(d) Sum of all divisors

$$=\left(\frac{3^5-1}{3-1}\right)\left(\frac{5^3-1}{5-1}\right)\left(\frac{7^4-1}{7-1}\right)$$
$$=\frac{(242)(124)(2400)}{2\times 4\times 6}$$
$=1500400$

6 Find exponent of

(a) 3 in 100! (b) 10 in 100! (c) Power of 5 in ${}^{30}C_{12}$

Solution:

(a) Exponent of 3 in 100!

$$=\left[\frac{100}{3}\right]+\left[\frac{100}{3^2}\right]+\left[\frac{100}{3^3}\right]+\left[\frac{100}{3^4}\right]+\left[\frac{100}{3^5}\right]+.....$$
$=33+11+3+1+0$
$=48$

(b) Exponent of 10 in 100!

$$=\left[\frac{100}{10}\right]+\left[\frac{100}{10^2}\right]+\left[\frac{100}{10^3}\right]+....$$
$=10+1+0$
$=11$

(c) ${}^{30}C_{12}=\dfrac{30!}{18!12!}$

$$\left[\frac{30}{5}\right]+\left[\frac{30}{25}\right]=7$$

$$\left[\frac{18}{5}\right]=3,\left[\frac{12}{5}\right]=2$$

Exponent of 5 = 7-3-2=2

7 The total number of selections of fruit which can be made from 3 bananas, 4 apples and 2 oranges is

Solution:

Total number of selection

= (3 + 1)(4 + 1)(2 + 1)

= 4 × 5 × 3

= 60

8 The number 2006 is made up of exactly two zeros and two other digits whose sum is 8. The number of 4 digit numbers with these properties (including 2006) is :

Solution:

let the other two digits a and b.

a + b = 8

(1,7), (2,6), (3,5), (4,4)

Number of digits = 3×2×3 + 3 = 21

9 Number of ways in which letters of the word ENGINEER can be arranged :

(a) so that the word starts with E but does not end with N.

(b) so that the word neither starts with E nor ends with N.

(c) so that vowels occur in alphabetical order.

(d) so that no two alike letters are together.

Solution:

(a) $\frac{7!}{2!2!}-\frac{6!}{2!}=900$

(b) Total – words either start with E or end with N

$$\frac{8!}{3!2!}-\left(\frac{7!}{2!2!}+\frac{7!}{3!}-\frac{6!}{2!}\right)$$

$$=6!\left(\frac{14}{3}-\frac{7}{4}-\frac{7}{6}+\frac{1}{2}\right)$$

$$=6!\left(\frac{112-42-28+12}{24}\right)$$

$$=6!\left(\frac{112-42-28+12}{24}\right)=\frac{720\times 9}{4}$$

$$=1620$$

(c) ${}^{8}C_{4}\times 1\times\frac{4!}{2!}=840$

(d) No alike letter together = No E together – (No E together and N together)

$$={}^{6}C_{3}\frac{5!}{2!}-{}^{5}C_{3}4!$$

= 1200-240 = 960

10 The number of divisors of $2^3 .3^3 .5^3 . 7^5$ of the form 4n + 1, n ∈ N is :

Solution:

$5^0, 5^1, 5^2, 5^3, 7^0, 7^1, 7^2, 7^3, 7^4, 7^5, 3^0, 3^1, 3^2, 3^3$ for factors of the form $4n+1$,

$(4k+1)(4k+1)(4k+1) \to 4\times 3\times 2 = 24$

and $(4k+3)(4k+3)(4k+1) \to 3\times 2\times 4 = 24$

$\Rightarrow$ total factors $= 24+24-1$

$= 47$

11 A disarranged number from the set 1 – 9 is an arrangement of all these number so that all numbers take up its unusual position. (e.g. 1 is in any place other than the first position, 2 is in any place other than the second position .... all the way to 9). Number of ways in which at least six number stake up their usual positions, is

Solution:

6 numbers take usual position + 7 number take + 9 number take.

Ways such that 6 numbers take usual

position = $\underbrace{{}^9C_3}_{\text{3 nos which don't take usual position}} \times \underbrace{2\times 1}_{\left(\text{Arranging those number first no. will have 2 options. Next 2 will have only1}\right)} = 168$

Ways such that 7 number take usual position = ${}^9C_2 = 36$

Ways such that 9 number take usual position = 1

Total ways = 168 + 36 + 1 = 205

It is interesting to note that only 8 numbers cannot take their usual positions.

12 How many ten digits whole number satisfy the following property they have 2 and 5 as digits, and there are no consecutive 2's in the number (i.e. any two 2's are separated by at least one 5).

Solution:

One 2, Two 2’s, Three 2’s,…., , Five 2’s

One 2 cases = 10

Two 2’s cases = ${}^9C_2$

Three 2’s cases = ${}^8C_3$

Four 2’s cases = ${}^7C_4$

Five 2’s cases = ${}^6C_5$

Total = 10 + 36 + 56 + 35 + 6 = 143

13 (a) In how many ways the number 7056 can be resolved as a product of 2 factors.

(b) Find the number of ways in which the number 300300 can be split into 2 factors which are relatively prime.

Solution:

$7056 = 2^4.3^2.7^3$

$$\frac{5\times3\times3+1}{2}=23$$

(b) $300300 = 2^2 .5^2 .3.11.13.7$

There are 6 primes.

Ways = 2^6 = 32

14 How many 4 digit numbers are there which contains not more than 2 different digits?

Solution:

aaaa type numbers = 9

aabb type numbers =Total – starting with zero

$$= {}^{10}C_2\frac{4!}{2!2!}-{}^{9}C_1\frac{3!}{2!}=270-27=243$$

abbb type numbers = Total – starting with zero

$$= {}^{10}C_2\,{}^{2}C_1\frac{4!}{3!}-{}^{9}C_1\left(\frac{3!}{2!}+1\right)=360-36=324$$

Total = 576

15 There are counters available in 7 different colours. Counters are all alike except for the colour and they are atleast ten of each colour. Find the number of ways in which an arrangement of 10 counters can be made. How many of these will have counters of each colour.

Solution:

Sol. (i) 7.7.......7 = 7^{10}

(ii) Different combinations are:

(a) 2, 2, 2, 1, 1, 1, 1

$${}^{7}C_3\frac{10!}{2!2!2!}=\frac{35}{8}10!$$

(b) 3, 2, 1, 1, 1, 1, 1

$${}^{7}C_2\,{}^{2}C_1\frac{10!}{3!2!}=\frac{7}{2}10!$$

(c) 4, 1, 1, 1, 1, 1, 1

$${}^{7}C_1\frac{10!}{4!}=\frac{7}{24}10!$$

Total = $\frac{35}{8}10!+\frac{7}{2}10!+\frac{7}{24}10!=\frac{105+84+7}{24}10!$

$$=\frac{196}{24}10!=\frac{49}{6}10!$$

16 In how many ways 6 letters can be placed in 6 envelopes such that

(a) No letter is placed in its corresponding envelope.

(b) At least 4 letters are placed in correct envelopes.

(c) At most 3 letters are placed in wrong envelopes

Solution:

Using of ways to place 6 letters in 6 envelop such that all are placed in wrong envelopes

$$\lfloor 6\left[1-\frac{1}{\lfloor 1}+\frac{1}{\lfloor 2}-\frac{1}{\lfloor 3}+....+\frac{1}{\lfloor 6}\right]$$

$$=360-120+30-6+1=265$$

Number of ways to place letters such that at least 4 letters are placed in correct envelopes

= 4 letter are placed in correct envelopes and 2 are in wrong + 5 letter are placed in correct envelops and 1 in wrong + All 6 letter are placed in correct envelopes

$= {}^{6}C_{4}\times 1+0$ (not possible to place in 1 in wrong envelop) +

$$1=\frac{6\times 5}{2}+1=16$$

(c) Number of ways to place 6 letters in 6 envelops such that a most 3 letters are placed in wrong envelopes = 0 letter is wrong envelop and 6 in correct + 1 letter wrong envelop and 5 in correct 2 letters in wrong envelopes and 4 are in correct + 3 letters in wrong envelopes and 3 in correct = 1+ 0 (not possible to place 1 in wrong envelop)

$$+\,{}^{6}C_{4}\times 1+{}^{6}C_{3}\ \lfloor 3\left[1-\frac{1}{\lfloor 1}+\frac{1}{\lfloor 2}-\frac{1}{\lfloor 3}\right]$$

$$=1+\frac{6\times 5}{2}+\frac{6\times 5\times 4}{6}\left(\frac{\lfloor 3}{\lfloor 2}-\frac{-\lfloor 3}{\lfloor 3}\right)$$

$$=1+15+20\times 2=56.$$

17 The number of seven digit integers, with sum of the digits equal to 10 and formed by using the digits 1,2 and 3 only, is

Solution:

There are two possible cases

Case I Five 1's, one 2's, one 3's

Number of numbers =7!/ 5!= 42

Case II Four 1's, three 2's

Number of numbers =7!/ 4!3! = 35

$\therefore$ Total Number of numbers = 42 +35 = 77

18 Let the product of all the divisors of 1440 be P. If P is divisible by 24^x, then the maximum value of x is

Solution:

$1440=2^5.3^2.5^1$

Number of divisors = $(5+1).(2+1).(1+1)=36$

Product of divisors = 1.2.3…480.720.1440.

Here all the 36 divisors ae written in the increasing order

They can be clubbed into 18 pairs, as shown below

(1.1440).(2.720).(3.480)…..etc

$\therefore$ Product of divisors = $(1440)^{18}=2^{90}.3^{36}.5^{18}$

$=\left(2^3.3\right)^{30}.3^6.5^{18}=24^{30}.3^6.5^{18}$ which is divisible by 24^x

$\therefore$ Maximum value of x = 30

19 The number of three digit numbers of the form xyz such that $x < y$ and $z \le y$ is

Solution:

If zero is included it will be at ${}^9C_2$ numbers

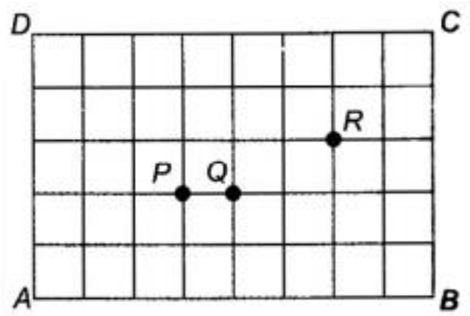
$$\text{If zero is excluded} \begin{cases} x, y, z \text{ all diff} & \Rightarrow \quad {}^9C_3 \times 21 \\ x = z < y & \Rightarrow \quad {}^9C_2 \\ x < y = z & \Rightarrow \quad {}^9C_2 \text{ numbers} \end{cases}$$

Total number of ways = 276

Alternate method:

y can be from 2 to 9 so total number of ways = $\sum_{r=2}^{9}\left(r^2 - 1\right) = 276$

20 Consider the network of equally spaced parallel lines(6 horizontal and 9 vertical) shown in the figure. All small squares are of the same size. A shortest route from A to Cis defines as a route consisting 8 horizontal steps and 5 vertical steps.



(a). The number of shortest routes through the junction P is:

(b) The number of shortest routes which go following street PQ must be

(c) The number of shortest routes which passes through junction P and R

Solution:

a)${}^5C_2 \times {}^8C_3 = 560$

(b) ${}^5C_2 \times {}^7C_3 = 350$

(c) ${}^5C_2 \times {}^4C_1 \times {}^4C_2 = 240$